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Competitive Coding

Experiment 4

Test Cases: ① Input $A = [1, 3, 5]$
Output $A = 8$

② Input $A = [2, 3]$
Output $A = 2$

We define $f(X, Y)$ as number of different corresponding bits in the binary representation of X and Y. For eg → $f(2, 7) = 2$, since the binary representation of 2 and 7 are 010 and 111, respectively. The first and the third bit differ, so $f(2, 7) = 2$.

You are given an array of N positive integers $A_1, A_2 \ldots A_N$. Find sum of $f(A_i, A_j)$ for all pairs (i, j) such that $1 \le i, j \le N$. Return the ans modulo $10^9 + 7$.

Solution → Given
$f(x, y) =$ no. of different bits in binary of x and y
[checking every pair of bits → $TC = (N^2)$ → TLE]
To solve this, we can use the concept of bit manipulation — By checking the bits of x and y

Lets look at a particular K^{th} bit
if K^{th} bit of x is 1 and K^{th} bit of y is 0
or vice versa
so ans = 2 for pair (x, y) and (y, x)

↳ Let cnt1 = nos having bit $K = 1$
cnt0 = nos having bit $K = 0$

contribution = 2 × cnt1 × cnt0

for (x, y) and (y, x)

↳ Steps to solve

For every bit (0 → 31)

1) Count numbers having that bit set.
2) Compute pairs
3) Add to answer.

Code :

```
# include <bits/stdc++.h>
using namespace std;

const long long MOD = 1e9 + 7;

long long totalPairs (vector<int>& A) {
    long long n = A.size();
    long long ans = 0;

    for (int bit = 0; bit < 32; bit ++) {
        long long cnt1 = 0;
        for (int i = 0; i < n; i++) {
            if (A[i] & (1 << bit)) {
                cnt1 ++;
            }
        }
        long long cnt0 = n - cnt1;

        long long Contribution = (cnt1 * cnt0) % MOD;
        contribution = (2* contribution) % MOD;

        ans = (ans + contribution) % MOD
    }

    return ans;
}
```

→ Dry Run

Input = {1, 3, 5}

1 → 001

3 → 011

5 → 101

for Bit 0 → 1(1), 3(1), (5(1)

⇒ cnt 1 = 0

cnt 0 = 0

Contribution = 2×3×0 = 0

for Bit 1 → 1(0), 3(1), 5(0)

⇒ cnt 1 = 1

cnt 0 = 2

Contribution = 2×1×2 = 4

for Bit 2 → 1(0), 3(0), ~~3~~5(1)

⇒ cnt 1 = 1

cnt 0 = 2

contribution = 2×1×2 = 4

Total ans = 0+4+4 = 8.

Pair	f(x,y)
(1,1)	0
(1,3)	1
(1,5)	1
(3,1)	1
(3,3)	0
(3,5)	2
(5,1)	1
(5,3)	2
(5,5)	0